

AION IO-LINK MIOTY ADAPTER PAYLOAD DESCRIPTION EN

Thank you for choosing the Aion sensor from Sentinum. Please read the following operating instructions carefully to prevent damage to the sensor, yourself, and the environment.



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1. DUTY CYCLE ENFORCEMENT

The sensor dynamically enforces the duty cycle itself based on the length of the data packets.

2. PAYLOAD DESCRIPTION

2.1. PAYLOAD LAYOUT

Byte indices in parentheses:

- (2-3) **VendorID**: UInt16 BE
- (5-6) **DeviceID**: UInt16 BE
- (8-9) **Measurement Value A**: device-specific (usually UInt16/Int16 × 0.1)
- (10-11) **Flags/status depending on device**
- (12-13) **Measurement value B**: device-specific
- (15) **Status nibble** (lower 4 bits) – **only** on some devices
- All other bytes (0-1, 4, 7, 14) Reserved.

Endianness: all 16-bit words are **big-endian** (readUInt16BE)

2.2. EXAMPLE PAYLOAD FOR LDH292

Byte index(es)	Length	Field	Data type	Scale	Unit	Notes
2-3	2	VendorID	UInt16 BE	×1	-	Expected: 0x0136 (dec. 310)
5-6	2	DeviceID	UInt16 BE	×1	-	Expected: 0x0539 (dec. 1337)
8-9	2	Humidity	UInt16 BE	×0.1	%RH	e.g., raw value 553 → 55.3 %RH
12-13	2	Temperature	UInt16 BE	×0.1	°C	e.g., raw value 234 → 23.4 °C
15 (low nibble)	4 bits	DeviceStatus	Enum	-	-	bytes[15] & 0x0F
-	-	(not used in the code)	-	-	-	Bytes 0-1, 4, 7, 10-11, 14 appear to be reserved/unclear

Status codes (lower 4 bits of byte 15)

Code	Meaning
0	OK
1	Maintenance required
2	Out of specification
3	Functional check
4	Failure
Other	Unknown (Code)

3. EXAMPLE LDH292

IDX	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
HEX	00	00	01	36	00	05	39	00	02	29	00	00	00	EA	00	00

- **Idx 2-3** → VendorID: 0x01 0x36 → 0x0136 → decimal **310**
- **Idx 5-6** → DeviceID: 0x05 0x39 → 0x0539 → decimal **1337**
- **Idx 8-9** → Humidity: 0x02 0x29 → 0x0229 → **553** × 0.1 → **55.3 %RH**
- **Idx 12-13** → Temperature: 0x00 0xEA → **234** × 0.1 → **23.4 °C**
- **Idx 15** → Status byte: 0x00; **low nibble** = 0x00 & 0x0F → **0 (OK)**